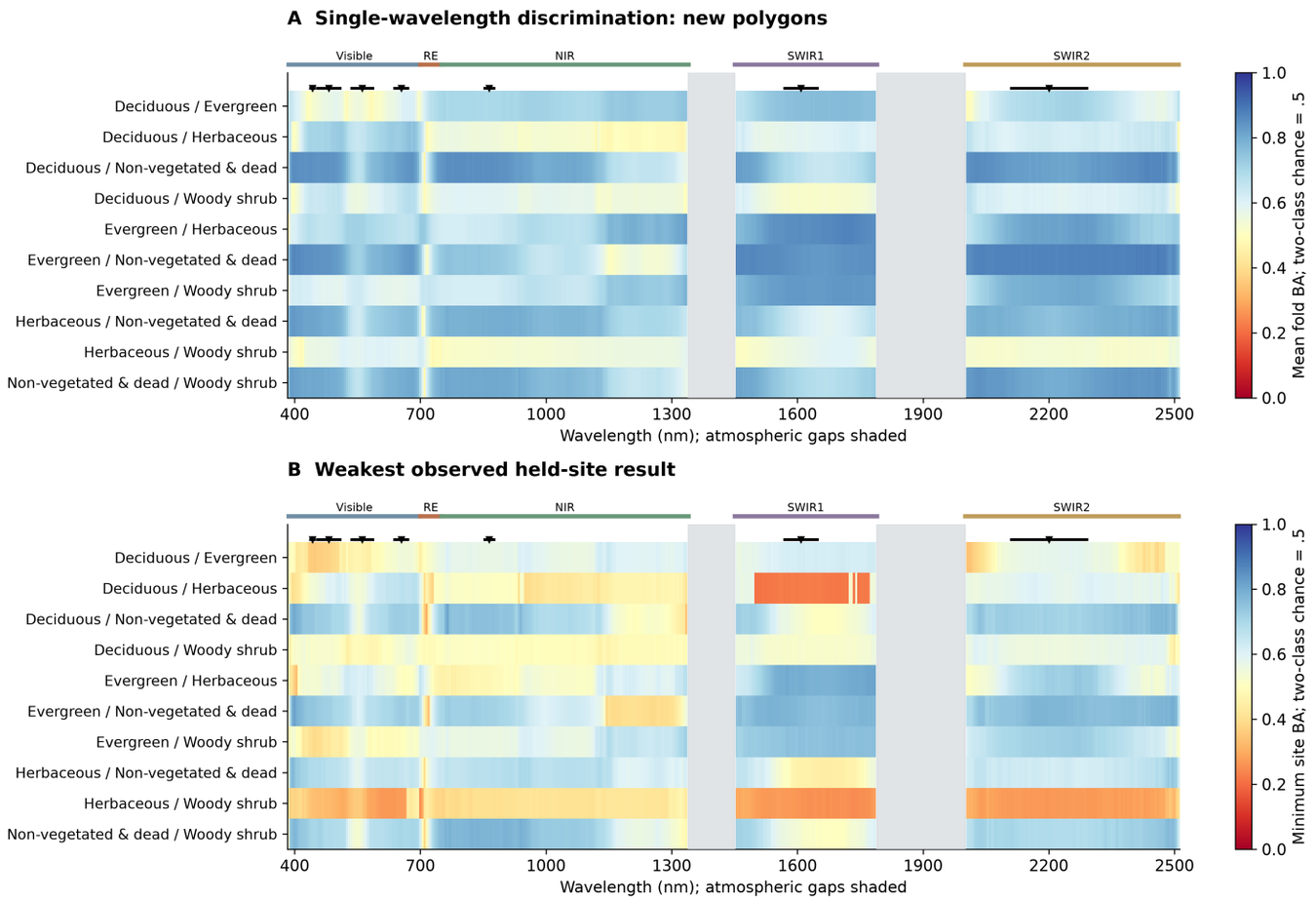
Researcher categories: where information lives

Wavelength discrimination, domain transfer and controlled synthetic abundance



One recorded label system

Deciduous; Evergreen; Herbaceous; Non-vegetated & dead; Woody shrub. Species-derived groupings are excluded from the current primary analysis.

The strongest new result

Supervised regression improves known synthetic fraction recovery: NEON ridge 0.216 versus MESMA 0.359 on new-parent tests, and 0.269 versus 0.362 on held sites. These gains do not establish unique inversion or field-cover accuracy.

A reusable information atlas

Ten category pairs ×360 supported NEON coordinates, with effects, adjusted evidence, prediction, stability, selection frequency and configured OLI support. Inputs remain in place; prior evidence is preserved.

3,639 polygon identities | 4,807 polygon/year spectra | NEON360 and synthetic OLI7

Fourteen main figures plus class-confusion supplement. Every figure explains its meaning and limits. A polygon is not necessarily one plant or a pure material.

Four questions require four different kinds of evidence

Do the sampled groups differ?

Whole-spectrum labels shuffle within acquisition-history blocks. Reuse forty full-cohort ensembles as descriptive evidence; separately, 230 training-only pair/cohort contrasts use 1,999 blocked shuffles, max-T and within-design Bonferroni correction. Neither p-map selects predictive features.

Can a new measurement be recognized?

Single-wavelength LDA learns means, direction and threshold from training identities. Balanced accuracy averages class recalls. Multivariate logistic selection and C tuning refit inside inner GlobalID folds. Entire IDs and their repeated years stay outside the outer training set.

How much spectral information is complementary?

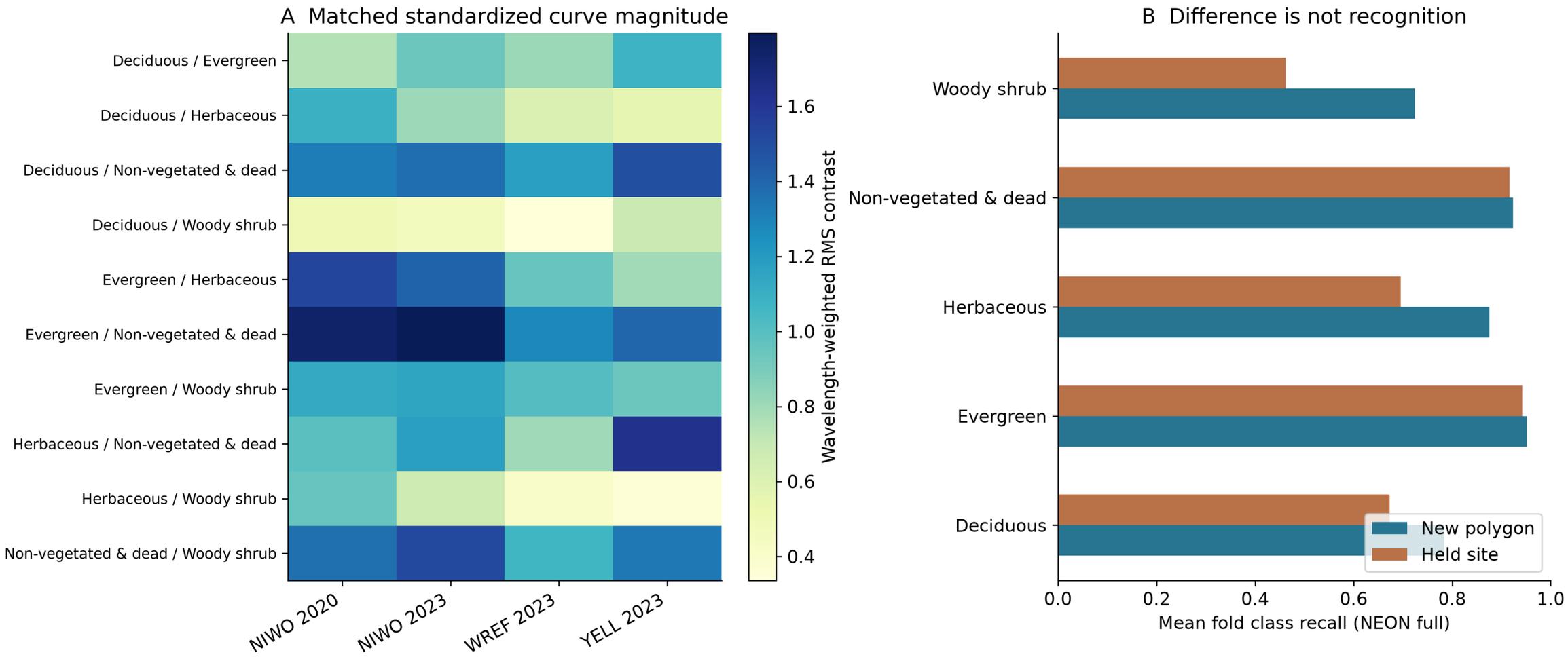
An orthogonal forward selector removes already-explained directions before choosing the next coordinate. Budgets span1 to360 coordinates and OLI7. Inner-only parsimony favors smaller budgets within two percentage points of the best inner score; outer results evaluate the policy.

Can fractions be recovered and transferred?

Every method receives the same controlled pair/triple mixtures. Proper/tuning/test parent identities are disjoint:120, 40 and up to 60 per experiment. Each five-parent bundle generates 170 dependent mixtures. Hyperparameters use tuning parents only; 499 uncertainty draws resample entire test bundles.

What the units mean: a GlobalID is a polygon footprint; neighboring wavelengths and generated mixtures are not independent biological replicates. Library units remain stored values. Synthetic OLI is a shared-source spectral representation. Annotation dates2023/2024 are reused across image years; paired tests assess retrospective footprints.

Figure 1. Whole-spectrum differences and recognition answer different questions



HOW TO READ IT

A: full-cohort acquisition-matched descriptive effects, not Cohen d. B: nested full-spectrum logistic class recall. Five recorded classes only; within-class spread and domain dependence remain.

WHAT WE LEARN

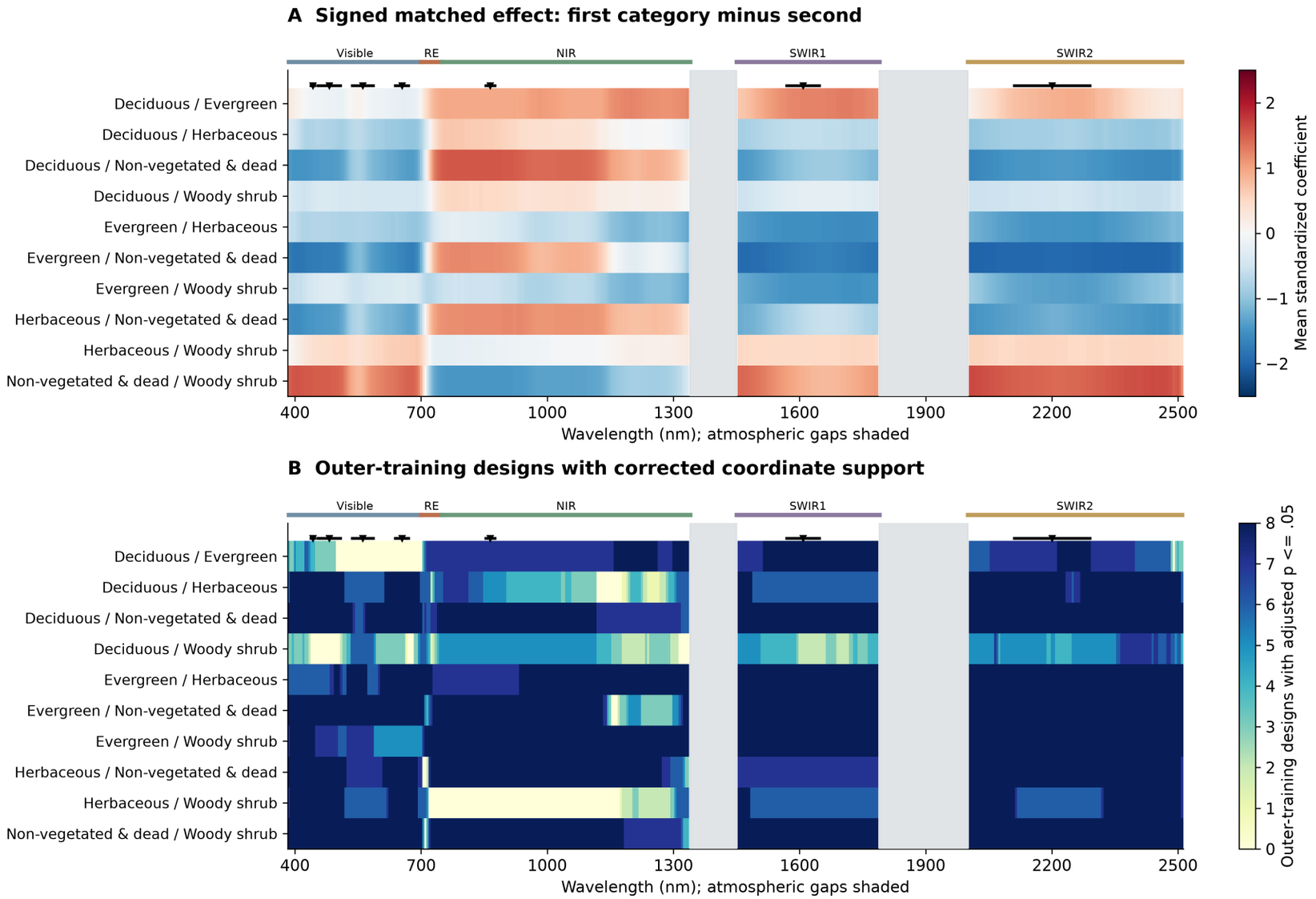
All forty eligible researcher-category whole-spectrum comparisons were supported in the prior matched NPC analysis. Magnitudes differ. In the new full-spectrum tests, mean held-site recall is 94.2% for Evergreen and 91.7% for Non-vegetated & dead, versus 46.3% for Woody shrub.

WHAT REMAINS UNCERTAIN

These are observational cover labels, not randomized traits or pure-material guarantees. Full-cohort matched effects and nested prediction use different populations/weighting.

Evidence: cohort_wavelength_inference.csv; class_specific_transfer.csv. Full paths, numerical products and hashes are retained in the versioned packet.

Figure 2. Where do the matched spectral differences occur?



HOW TO READ IT

A: full-cohort descriptive effects. B: training-only max-T and within-design Bonferroni support in eight outer-training designs, which overlap and are not independent replications. No inference map selects predictive features. Triangles/short bars mark configured OLI centers/FWHM. Visible <700; red edge 700-750; NIR 750-1340; SWIR1 1450-1790; SWIR2 2000-2510 nm.

WHAT WE LEARN

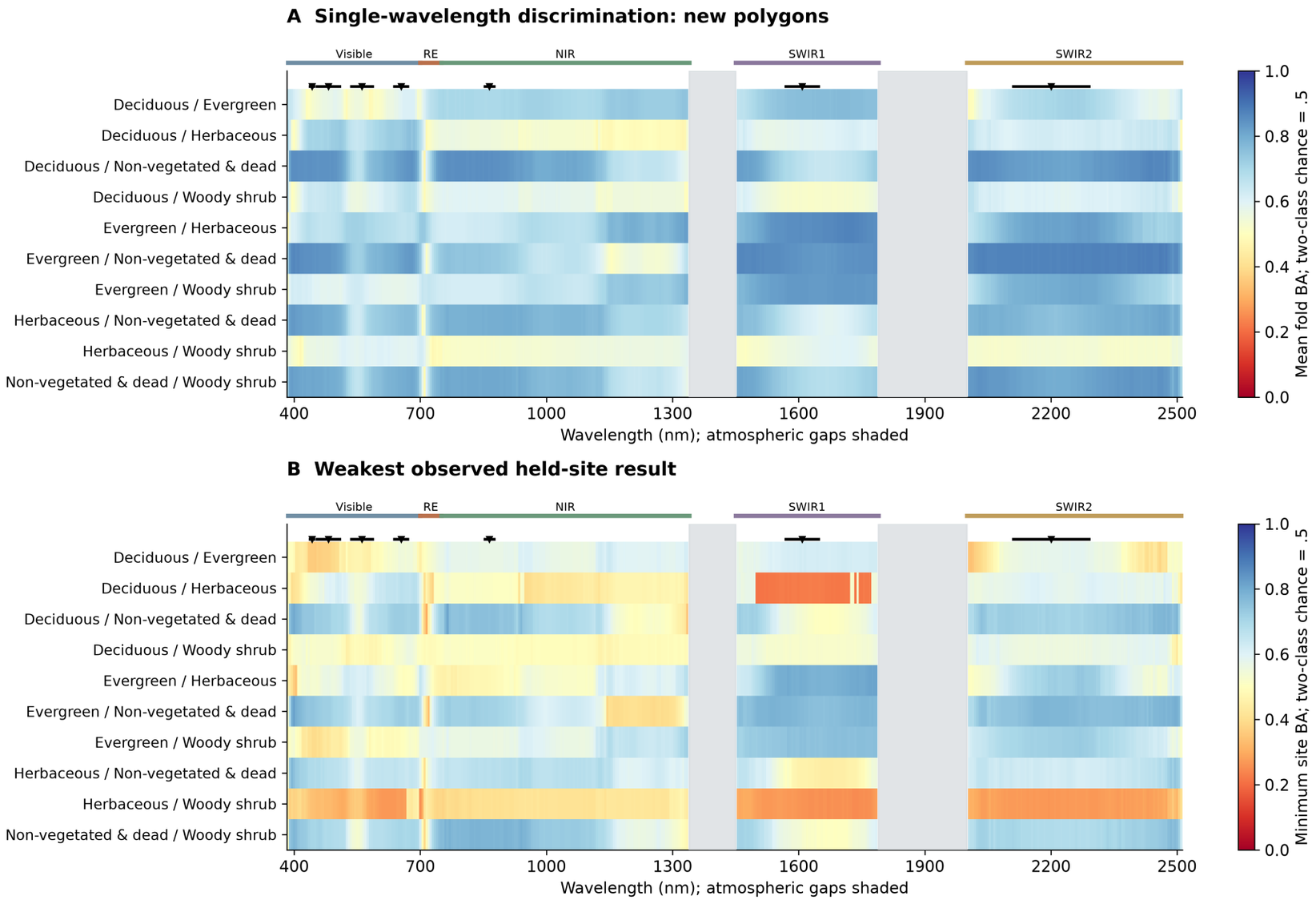
Strong differences occupy broad regions rather than isolated independent channels. Training-only corrected support is shown across eight overlapping outer-training designs; these are not independent replications. The signed mean can conceal reversals; consult individual site/year effects and sign consistency in the atlas.

WHAT REMAINS UNCERTAIN

Full-cohort evidence and training-only evidence are separate. Max-T and within-design Bonferroni control the declared training family. Neither inference map selects predictive features. Null support is not an effect confidence interval.

Evidence: researcher_pft_information_atlas.csv. Full paths, numerical products and hashes are retained in the versioned packet.

Figure 3. Which individual wavelengths predict held-out categories?



HOW TO READ IT

Balanced-prior univariate LDA; direction and threshold trained without target identities. Mean versus weakest-site performance are different views. These are estimates for one rule, not a theoretical information bound.

WHAT WE LEARN

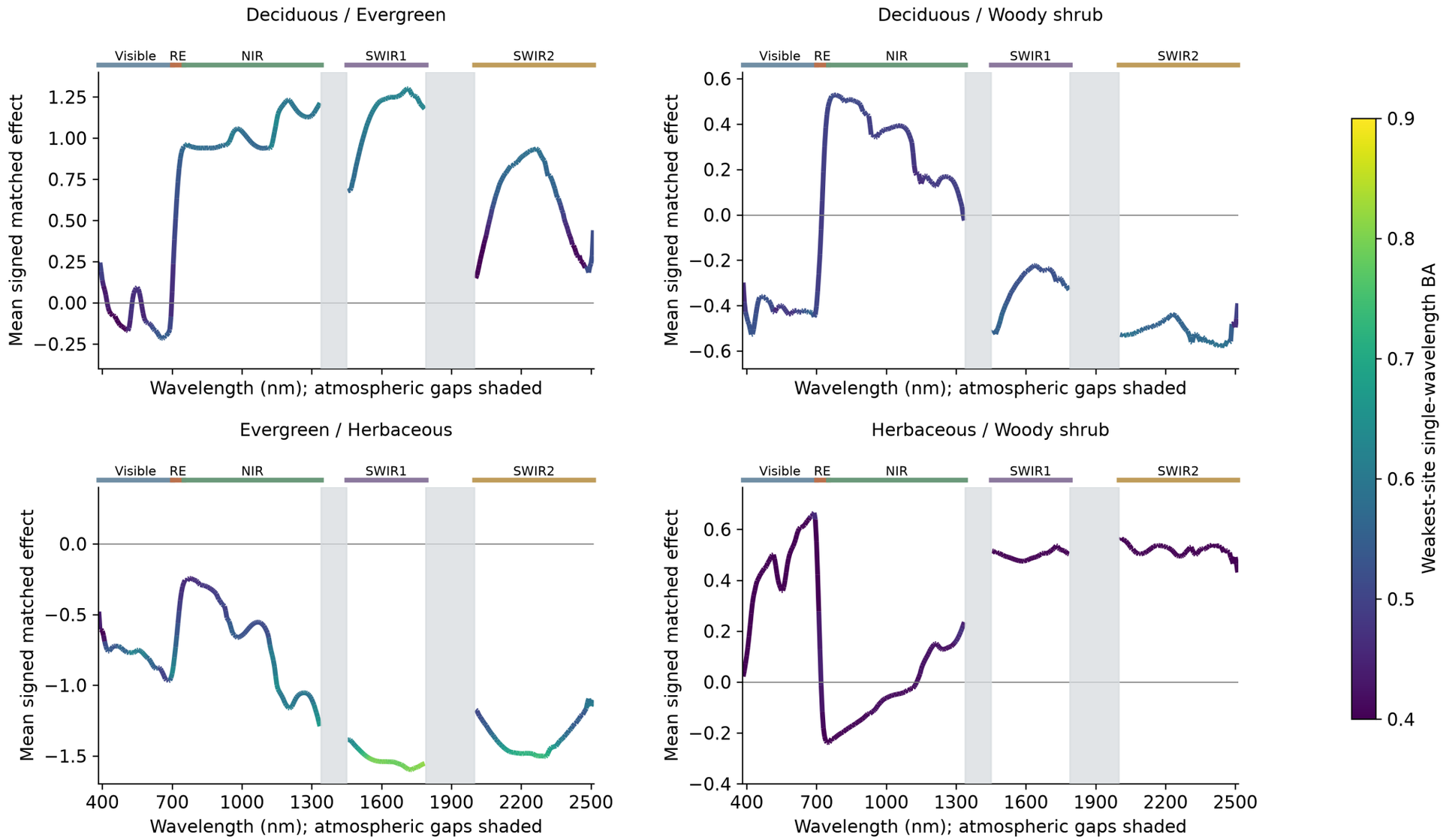
Evergreen/Herbaceous and several contrasts involving Non-vegetated & dead have stronger single-coordinate signals. Herbaceous/Woody shrub remains below chance at its best observed worst-site coordinate under this mean-based rule.

WHAT REMAINS UNCERTAIN

LDA learns means, direction and midpoint threshold on training IDs only. It estimates one rule, not an upper bound on all nonlinear single-coordinate information. Outer peaks are descriptive, not selected deployment bands.

Evidence: single_wavelength_discrimination.parquet; researcher_pft_information_atlas.csv. Full paths, numerical products and hashes are retained in the versioned packet.

Figure 4. A large group difference can still be a weak predictor



HOW TO READ IT

Four declared illustrations; all ten pairs are in the atlas. Effect height describes matched group differences; color describes a separate held-site predictor. Segments do not cross atmospheric gaps.

WHAT WE LEARN

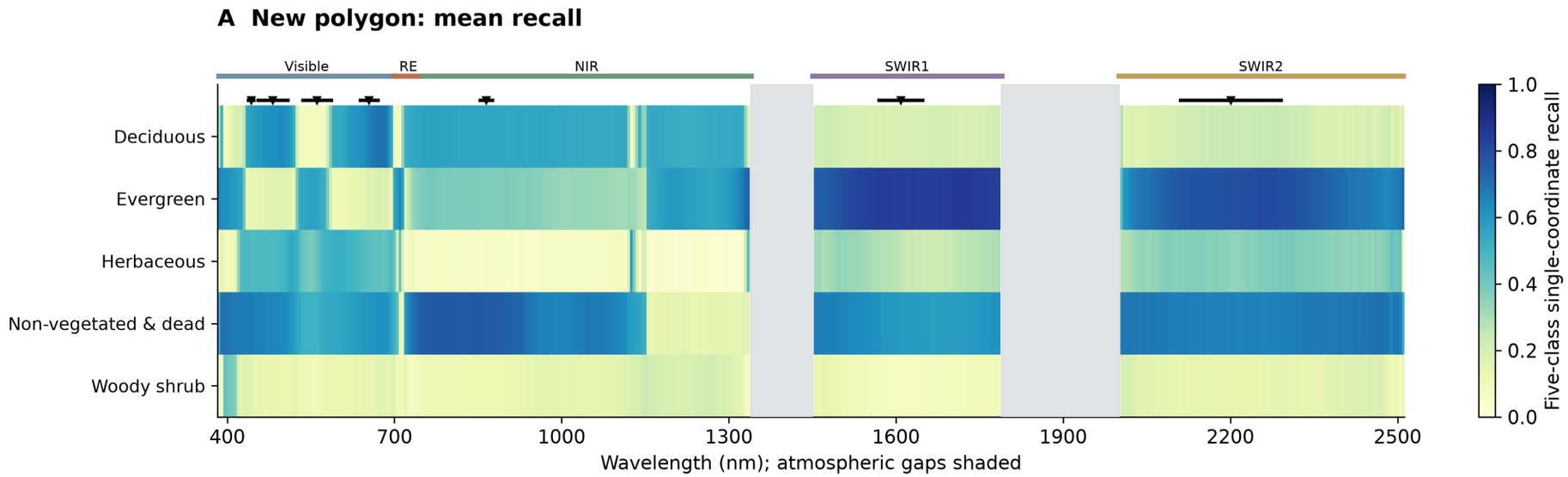
These four selected pairs compare two distinct quantities: adjusted curve magnitude and held-site prediction. Effect height is not accuracy. Near-infrared and shortwave-infrared information helps some pairs, while the difficult Herbaceous/Woody shrub contrast lacks a robust single-coordinate rule.

WHAT REMAINS UNCERTAIN

Illustrations are selected for interpretation, not another testing family. All ten pairs are available in the full map. Spectral association does not uniquely identify a biochemical cause.

Evidence: researcher_pft_information_atlas.csv. Full paths, numerical products and hashes are retained in the versioned packet.

Figure 5. Which classes can one spectral coordinate recognize?



HOW TO READ IT

Prespecified five-class balanced-prior nearest-mean discriminant. A single coordinate can separate two classes yet confuse the five-way classification. This differs from pairwise balanced accuracy and full-spectrum logistic recall.

WHAT WE LEARN

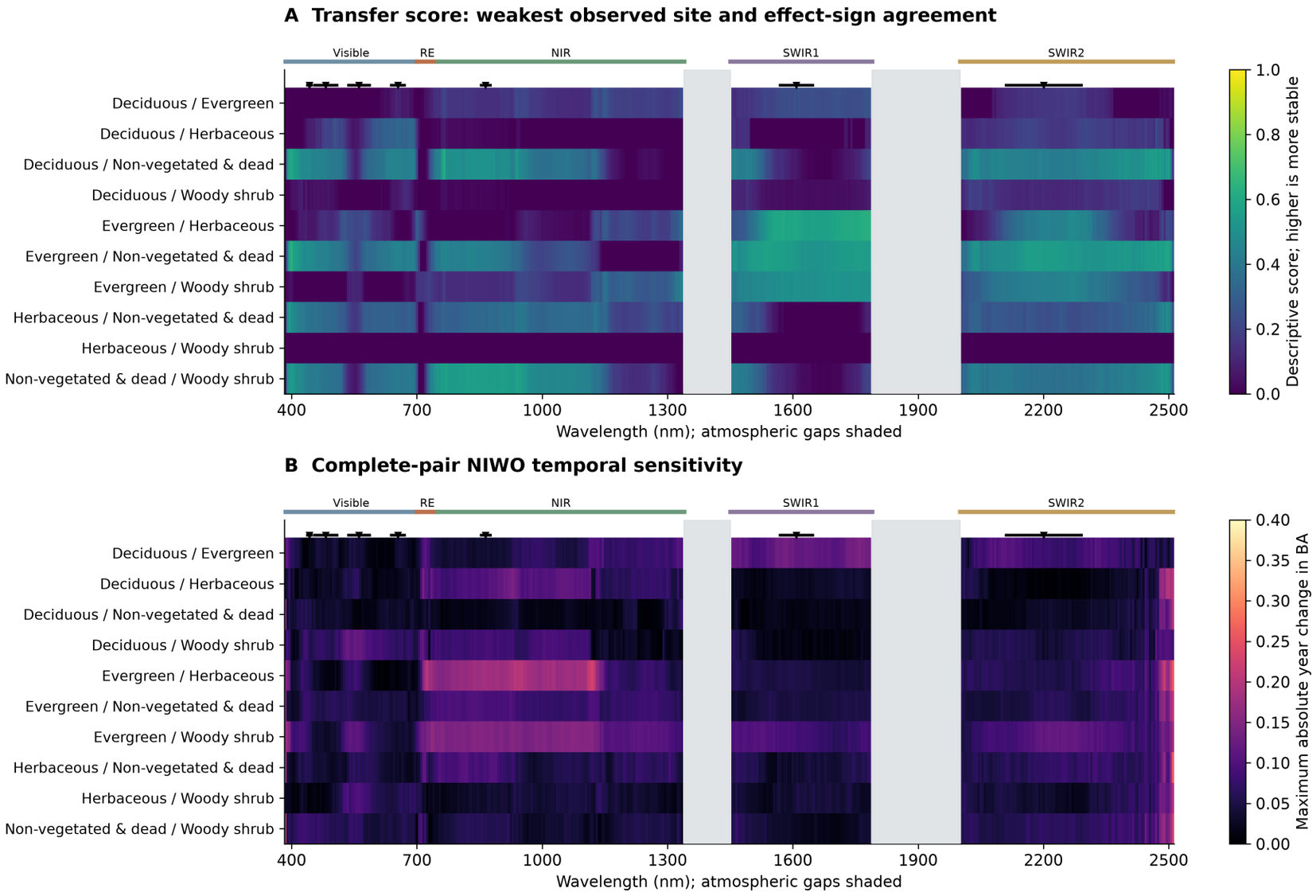
A coordinate that helps a two-class comparison need not distinguish all five classes at once. Mean-based class centers can crowd together. Full-spectrum logistic recognition is a separate, stronger model; these are not its wavelength importances.

WHAT REMAINS UNCERTAIN

Five-class nearest-mean recall differs from pairwise balanced accuracy. Class sizes differ; every identity, including its repeated years, stays together in train/test splitting.

Evidence: single_wavelength_class_recall.csv. Full paths, numerical products and hashes are retained in the versioned packet.

Figure 6. Separate spectral magnitude from site and year stability



HOW TO READ IT

Score=max(0,2*minimum held-site BA-1) times cohort sign consistency. It is a review aid over three observed sites. Temporal changes preserve target identities; annotations are reused later field labels. No geographic-population or annual-truth guarantee.

WHAT WE LEARN

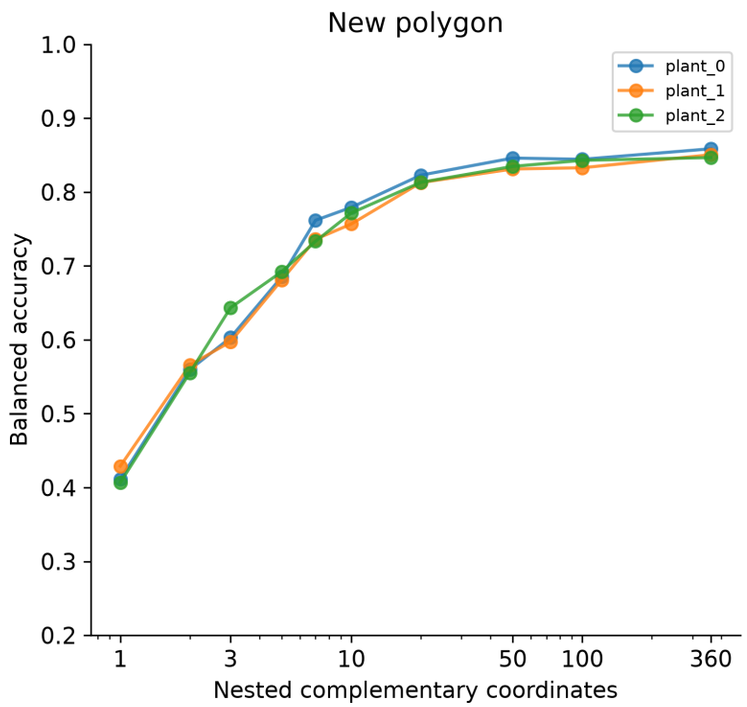
Some regions retain sign and predictive performance across sites; others are geographically specific or change between the paired image years. The score rewards the weakest observed site, so averaging cannot hide a failed domain.

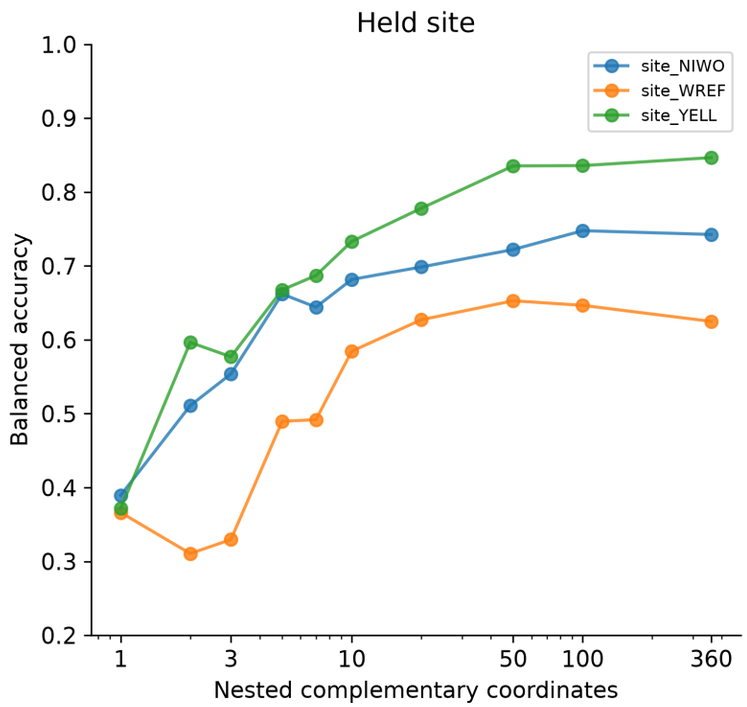
WHAT REMAINS UNCERTAIN

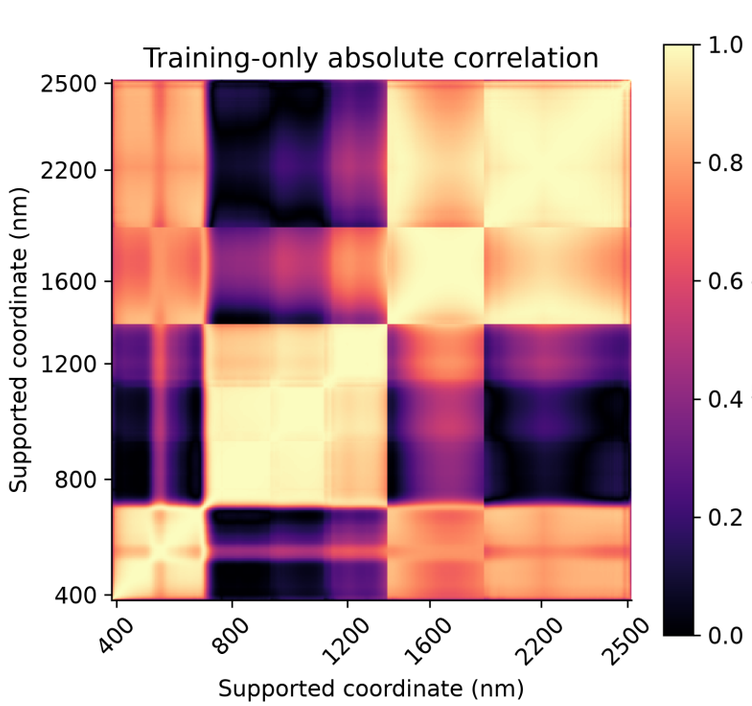
Only three sites and two incompletely crossed years are available. Labels dated 2023/2024 are reused across image years. These are retrospective footprints, not independently verified annual individuals.

Evidence: researcher_pft_information_atlas.csv; domain_rank_consistency.csv. Full paths, numerical products and hashes are retained in the versioned packet.

Figure 7. Complementary information accumulates over many coordinates







HOW TO READ IT

All selection and C tuning refit inside group-aware inner folds. Correlation panel compresses unsupported gaps in its coordinate index and interpolates no missing channel. The greedy residual criterion seeks complementary information; it is not an optimized sensor.

WHAT WE LEARN

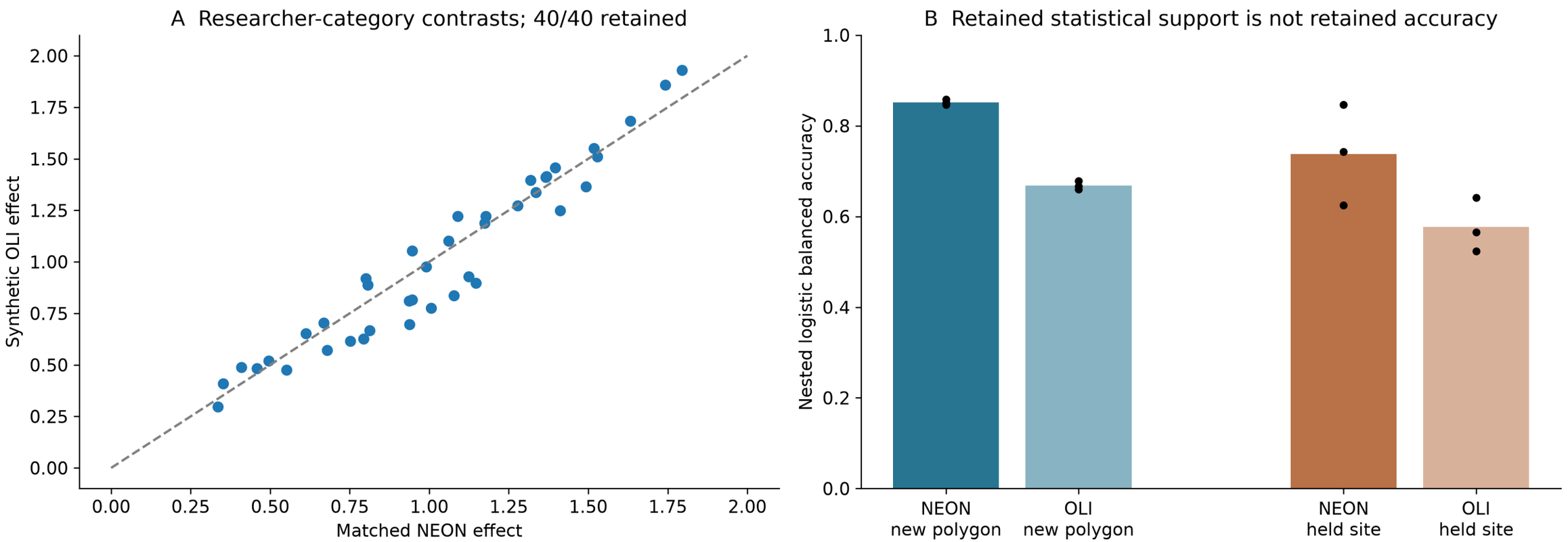
New-polygon balanced accuracy rises from 41.6% at one complementary coordinate to 81.6% at twenty and 85.2% for the full grid. Held-site values are 70.1% and 73.8%. The inner-only parsimony rule chooses 50-100 coordinates across these folds.

WHAT REMAINS UNCERTAIN

Selectors and regularization refit inside each inner identity fold. The greedy residual path is not a globally optimal sensor. Correlation alone does not establish conditional predictive redundancy.

Evidence: complementary_feature_performance.csv; complementary_selection.csv; training_correlation_clusters.csv. Full paths, numerical products and hashes are retained in the versioned packet.

Figure 8. Synthetic OLI preserves tests better than predictive accuracy



HOW TO READ IT

Identical class labels and paired source observations. OLI spectral convolution is synthetic, at native/polygon support; it does not supply independent sensor noise, a 30 m footprint or field validation.

WHAT WE LEARN

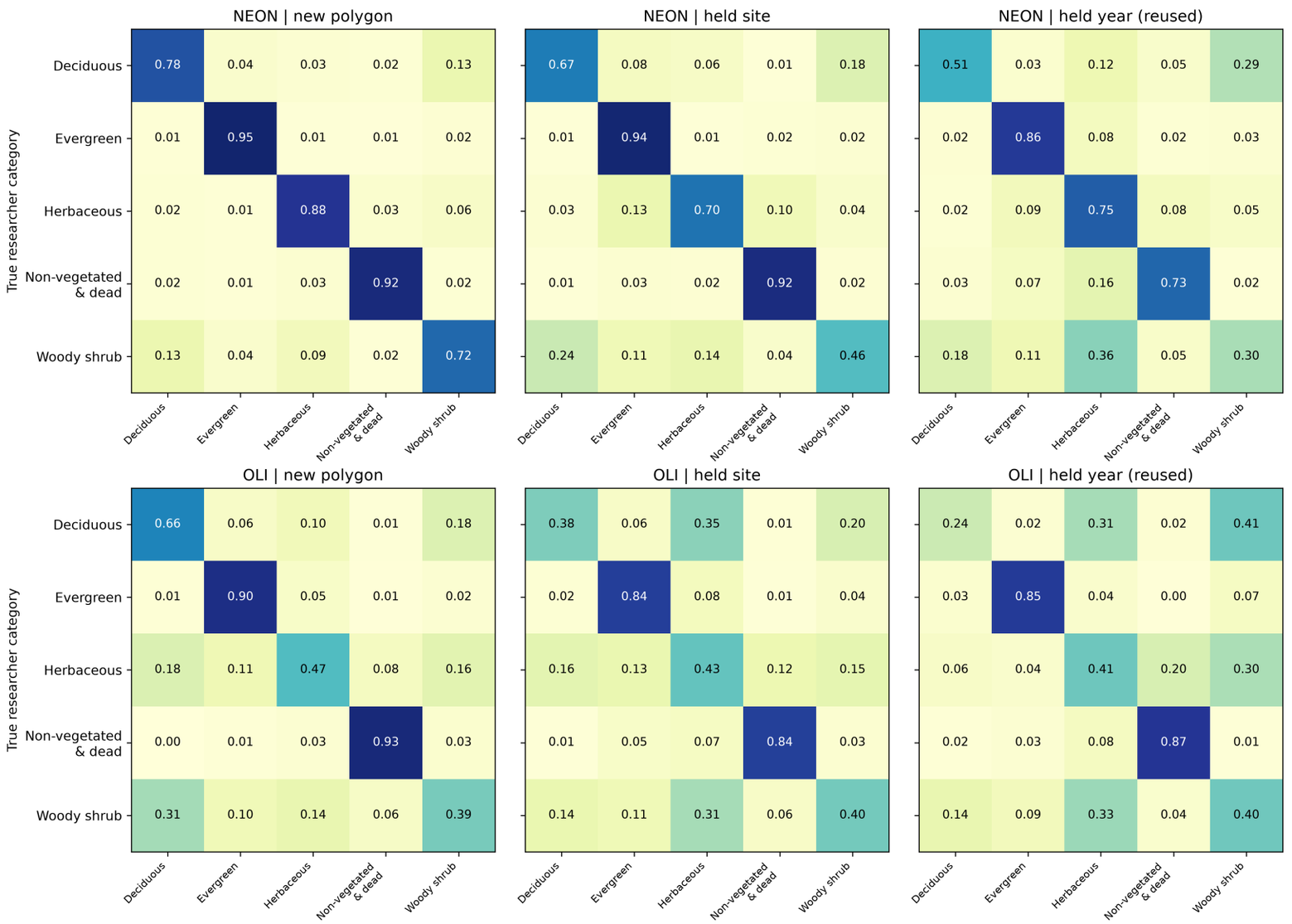
All forty researcher contrasts retain statistical support in OLI. Yet full-spectrum new-polygon/held-site accuracy changes from 85.2%/73.8% for NEON to 66.8%/57.7% for synthetic OLI. Support retention is not preservation of all predictive information.

WHAT REMAINS UNCERTAIN

OLI is derived from the same NEON spectra at native/polygon support. It is neither independent Landsat validation nor 30 m spatial aggregation. MicaSense is excluded because physical coordinates remain unresolved.

Evidence: complementary_feature_performance.csv; reused_researcher_sensor_effect_retention.csv. Full paths, numerical products and hashes are retained in the versioned packet.

Figure S1. Which researcher categories are confused?



HOW TO READ IT

Rows are true classes; columns are predictions. Cells are fold-mean row fractions. The year panels reuse the earlier validated researcher-only logistic baseline and annual-row target weighting; new polygon/site panels use the new nested identity-weighted model. These are conditional observed-domain tests.

WHAT WE LEARN

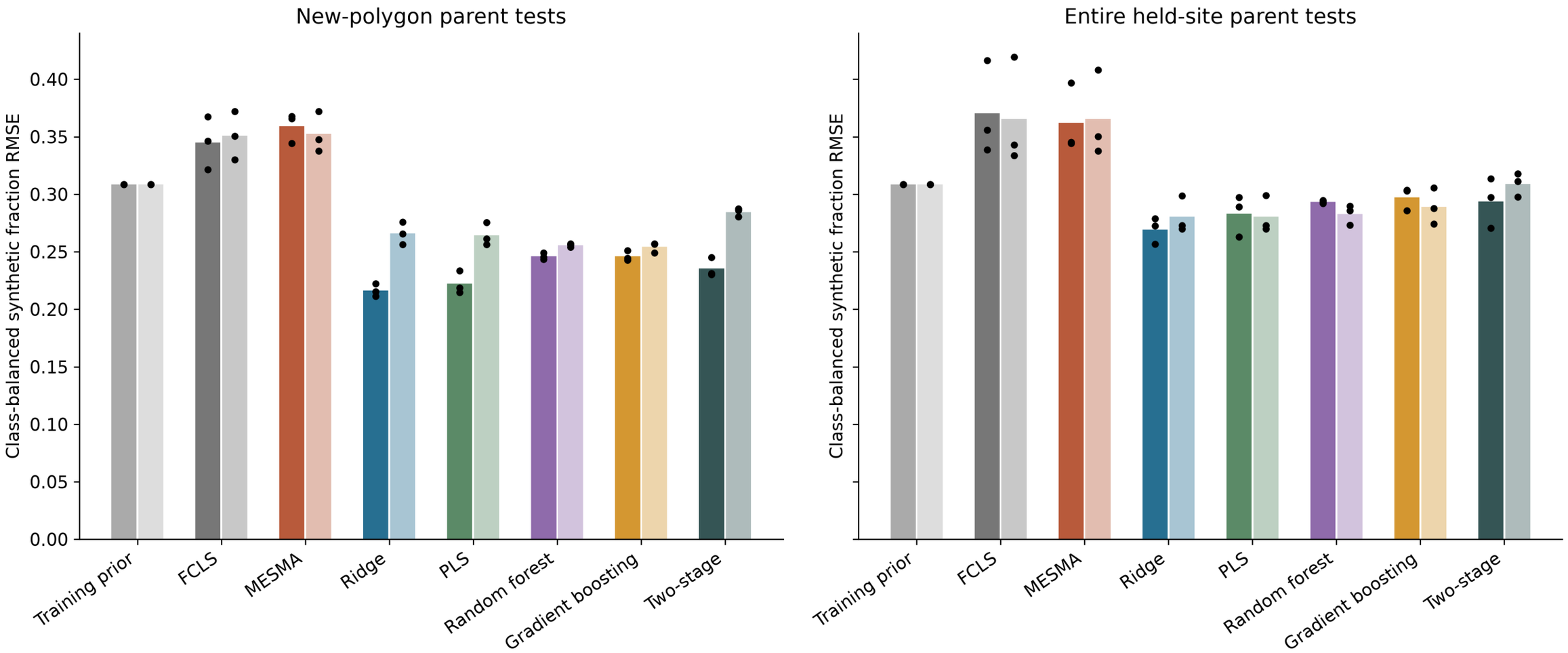
Evergreen and the broad background category transfer better than Woody shrub. Off-diagonal cells identify the alternatives receiving mistakes. The year panels preserve the prior researcher-only model and clearly label that reuse.

WHAT REMAINS UNCERTAIN

Rows are normalized within each fold, then averaged. The new polygon/site model uses identity weighting; historical year panels use the validated earlier annual-row model. Do not interpret these as identical fitted pipelines.

Evidence: confusion_matrices.csv; reused_researcher_year_confusion.csv. Full paths, numerical products and hashes are retained in the versioned packet.

Figure 9. Supervised models improve known synthetic fraction recovery



HOW TO READ IT

Solid bars: NEON; pale bars: OLI. Dots are outer folds, not independent fraction levels. All methods use identical parent-disjoint targets; 120 proper and 40 tuning parents per fold, up to 60 test parents. These are synthetic weights, not field cover.

WHAT WE LEARN

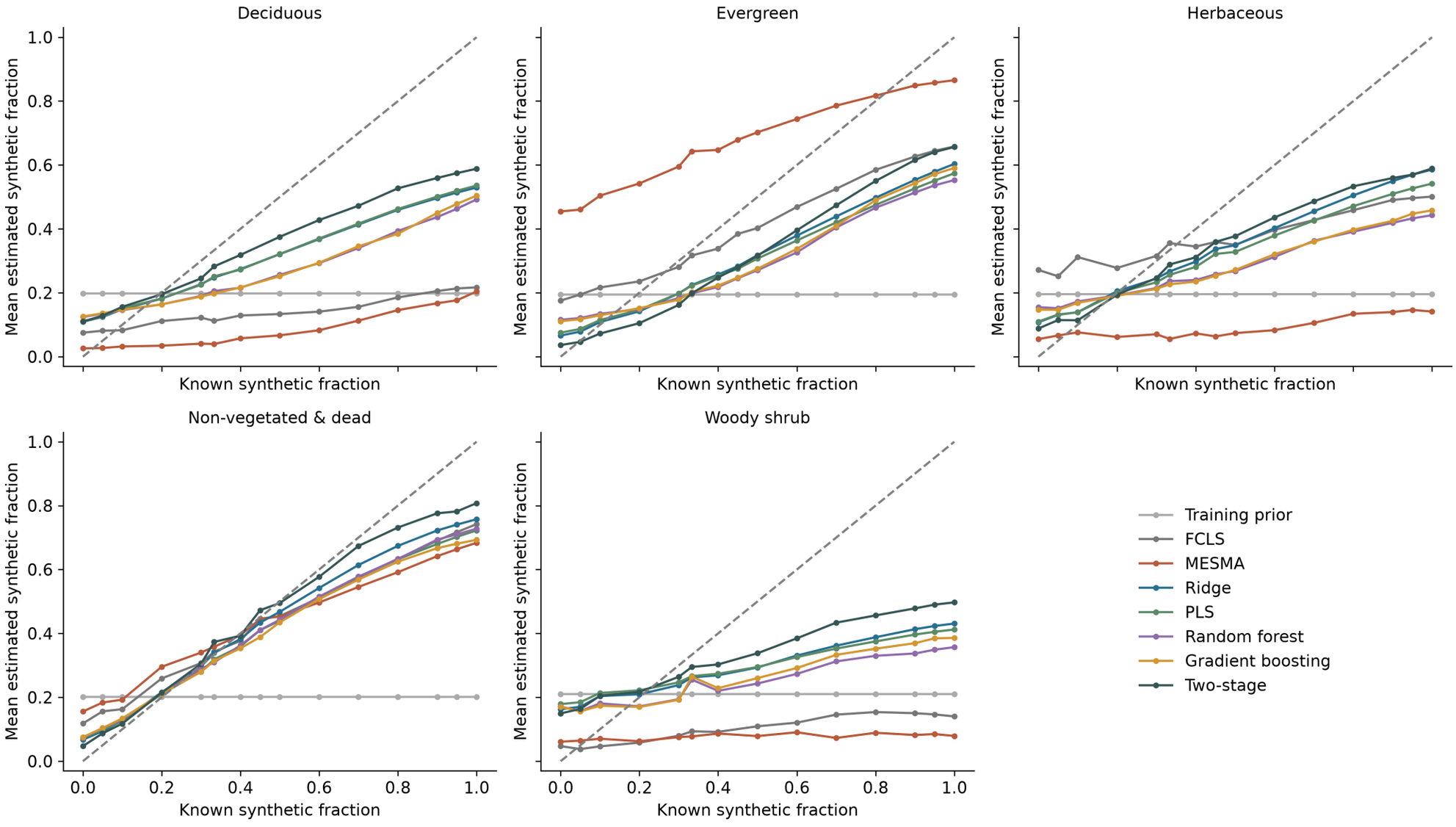
NEON ridge fraction RMSE is 0.216 for new-polygon parents and 0.269 for held sites, versus MESMA 0.359/0.362. The no-spectrum training-prior control is 0.309: ridge beats it for new-parent/held-site targets, but not on average in the global-year tests. Flexible trees do not outperform ridge in the principal mean.

WHAT REMAINS UNCERTAIN

Each experiment uses 120 proper and 40 tuning parents, with up to 60 test parents. All methods share targets. Synthetic rows within a parent bundle are dependent; field abundance is not measured.

Evidence: abundance_model_comparison.parquet; abundance_paired_uncertainty.csv. Full paths, numerical products and hashes are retained in the versioned packet.

Figure 10. Fraction errors depend on both class and true abundance



HOW TO READ IT

Curves average the three new-polygon folds; complete grids for all methods/domains are saved. At true zero, positive detection is false detection. The .10 cutoff is declared, not a universal detection limit. Mixtures sharing parents are dependent.

WHAT WE LEARN

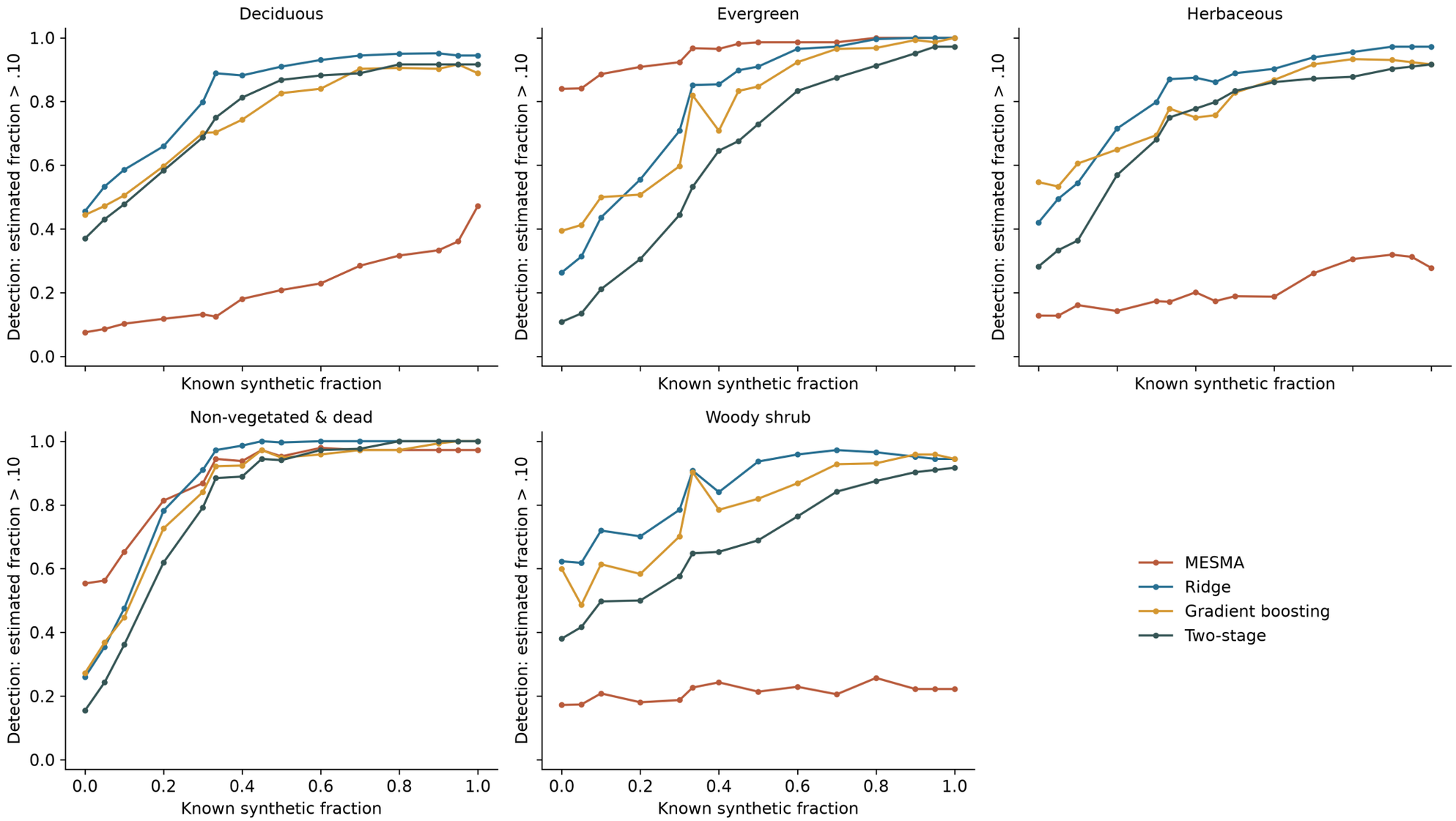
Models can shrink extremes, assign nonzero fractions to absent classes and differ substantially among researcher categories. Overall RMSE hides these patterns. The common projection enforces physical fraction constraints but cannot certify the biological interpretation.

WHAT REMAINS UNCERTAIN

Known fractions are algebraic source weights, not independent physical cover. Curves average folds on the stated synthetic prior. Every saved fraction is nonnegative and sums to one after projection.

Evidence: abundance_by_true_fraction.csv. Full paths, numerical products and hashes are retained in the versioned packet.

Figure 11. Fewer false detections can mean more missed low fractions



HOW TO READ IT

Curves average the three new-polygon folds; complete grids for all methods/domains are saved. At true zero, positive detection is false detection. The .10 cutoff is declared, not a universal detection limit. Mixtures sharing parents are dependent.

WHAT WE LEARN

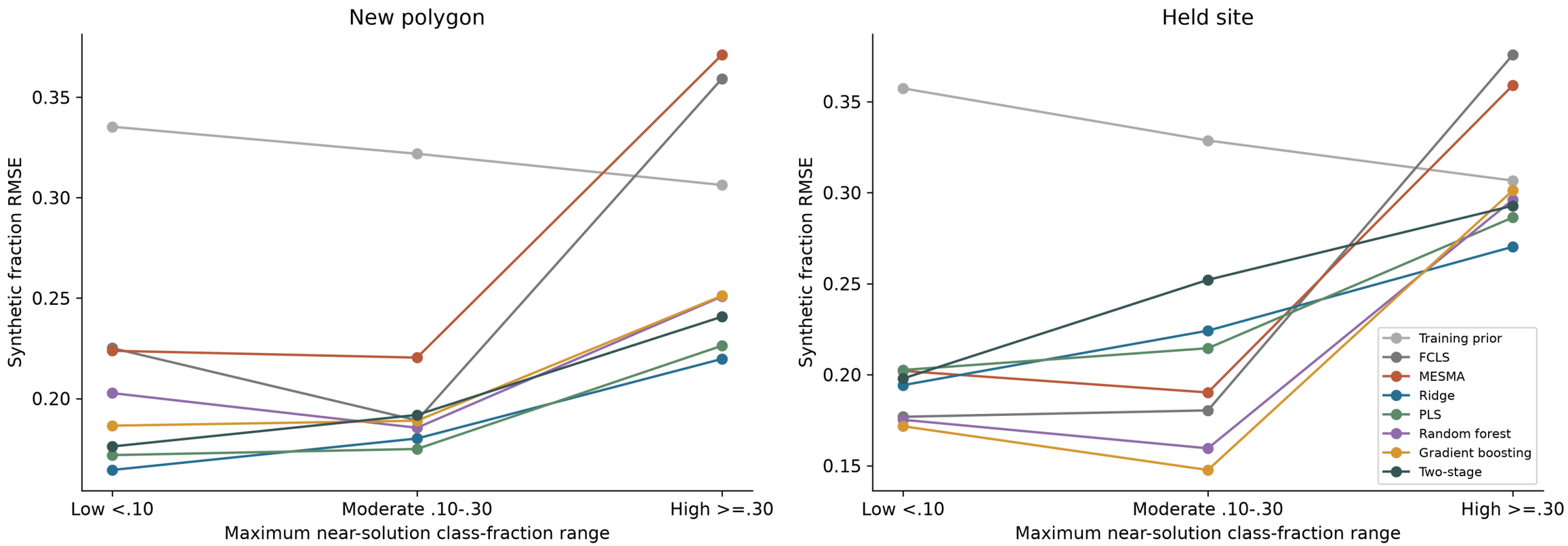
For Evergreen in new-parent tests, two-stage modeling reduces zero-fraction false detections relative to ridge, but increases low-fraction misses. Other classes show their own tradeoffs. A lower average fraction RMSE does not establish a reliable presence detector.

WHAT REMAINS UNCERTAIN

The .10 fraction cutoff is a stated comparison rule, not a universal detection limit. Parent-bundle dependence limits uncertainty; source class truth does not certify pure endmembers.

Evidence: abundance_by_true_fraction.csv. Full paths, numerical products and hashes are retained in the versioned packet.

Figure 12. Lower fraction error does not remove spectral ambiguity



HOW TO READ IT

Strata are shared across every method, using full-NEON MESMA near solutions within .01 training-RMS-scaled tolerance. Missing strata are unsupported. This diagnoses a candidate-library inverse problem, not intrinsic physical ambiguity or calibrated intervals.

WHAT WE LEARN

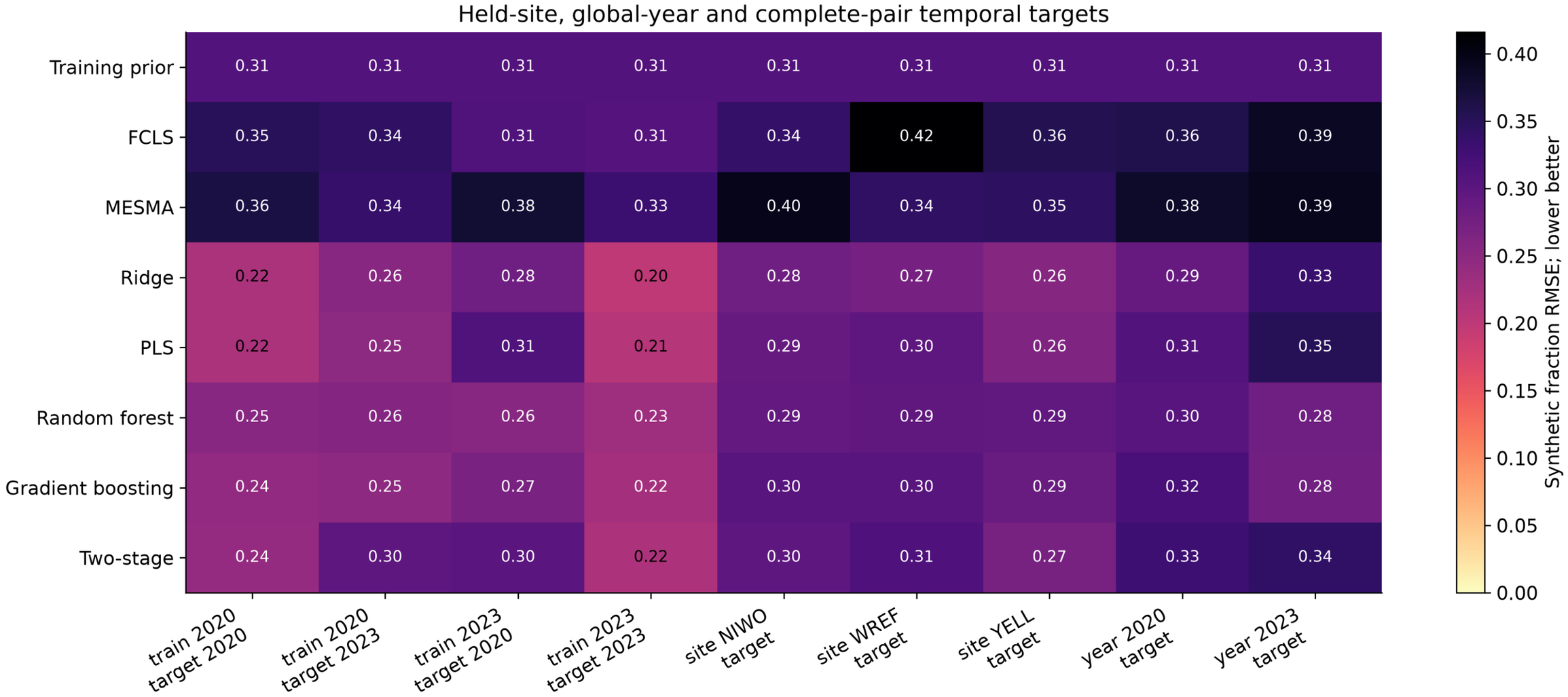
Ridge improves average error in the high-ambiguity held-site stratum, but gains are not uniform: MESMA can perform better in a moderate stratum. A learned prior can improve expected error while another biological composition still fits almost as well.

WHAT REMAINS UNCERTAIN

Ambiguity is candidate-library- and tolerance-dependent. Strata use the same NEON diagnostic for all methods. Missing strata are unsupported, and near ranges are not calibrated confidence intervals.

Evidence: abundance_by_ambiguity.csv. Full paths, numerical products and hashes are retained in the versioned packet.

Figure 13. Abundance transfer depends on the target domain



HOW TO READ IT

Global year holds alter site composition. Paired NIWO directions reuse the same held-out target IDs/fractions in both years, with other IDs for proper/tuning pools. Annotation labels are retrospective. Each cell is conditional on sampled parent bundles.

WHAT WE LEARN

Held-site and year targets alter model performance. Ridge global-year mean RMSE is 0.311, versus 0.309 for the no-spectrum training prior. The complete-pair NIWO design tests the same unseen IDs/fractions in both years, using other identities for fitting. Global year holds additionally change site composition; those columns are not a clean causal temporal effect.

WHAT REMAINS UNCERTAIN

Paired abundance is a fixed identity split, not the three-fold single-wavelength temporal experiment. Interval estimates condition on observed parent bundles; they do not cover new geographic populations.

Evidence: abundance_model_comparison.parquet. Full paths, numerical products and hashes are retained in the versioned packet.

Figure 14. The final evidence is conditional, with a field-truth gate

Question	Measured result	Evidence status
Five-category recognition: plant	NEON 85.2%; synthetic OLI 66.8%	Conditional evidence
Five-category recognition: site	NEON 73.8%; synthetic OLI 57.7%	Conditional evidence
Synthetic fraction recovery: plant	MESMA 0.359; lowest observed Ridge 0.216	Conditional synthetic evidence
Synthetic fraction recovery: site	MESMA 0.362; lowest observed Ridge 0.269	Conditional synthetic evidence
Independent field abundance	No independent complete fraction truth supplied	Not established

HOW TO READ IT

Evidence status is use-specific, not an aggregate score. Lowest observed outer score is descriptive, not a tuning choice or multiplicity-corrected winner. No synthetic success is promoted to field abundance validation.

WHAT WE LEARN

Researcher-category recognition and controlled synthetic fraction recovery are useful, with clear domain limits. Supervised regression improves the latter. Unique physical inversion and operational fractional cover remain unestablished without independent purity and complete field fractions.

WHAT REMAINS UNCERTAIN

The lowest observed outer score is a descriptive comparison, not test-set hyperparameter tuning or a multiplicity-corrected winner. No current result is promoted to field abundance validation.

Evidence: final_evidence_summary.csv. Full paths, numerical products and hashes are retained in the versioned packet.

What improves, what fails, and what the data cannot decide

Recognition is class- and domain-dependent

Evergreen and the broad background category transfer more consistently than Woody shrub. Whole-spectrum support does not guarantee useful single-coordinate prediction. Exact wavelengths and all domain values are in the atlas; observed peaks are not deployed sensor recommendations.

Better average fractions are not unique inversion

ML can exploit relationships in the finite synthetic training distribution. High-ambiguity targets can still show reduced average error, without excluding another near-fitting composition. Gains vary by stratum and class; modest-ambiguity cases do not uniformly favor ML.

Complementarity is real, but not a sensor optimum

A single coordinate performs poorly on the five-way task. Multiple residual-selected measurements improve recognition; the inner-only policy chooses50-100 coordinates in these folds. Correlation clusters explain redundancy, but no universal band count follows from three sites.

Detection and abundance have different costs

Two-stage gating reduces some absent-class predictions but increases missed low fractions. Shrinkage, false positives, calibration and low-fraction error must be read together. A .10 cutoff and conditional curves do not supply a universal detection limit.

Simple supervised models earn their place

Projected ridge improves known synthetic fractions under new-polygon and held-site tests and beats a no-spectrum training-prior control. Its global-year mean does not beat that prior; gains over MESMA alone are insufficient. Trees do not improve the principal average. First/second derivatives are worse in this benchmark; other transformations remain reported rather than silently selected.

Failure cannot be causally assigned to one factor

Near-zero known-parent oracle error shows that the solver can recover the idealized linear weights. The remaining gap implicates representation/library variability and model choice, with domain and ambiguity effects. Missing independent purity/fraction truth still blocks a field claim.

The evidence needed before a field-abundance claim

Recover physical provenance first

Verify original reflectance units/scaling, physical band headers and polygon purity. OLI export coordinates are rounded labels; response annotations use the exact configured parameters. MicaSense physical coordinates remain unresolved and are excluded. Non-vegetated & dead is a heterogeneous cover class.

Measure independent fractions, absences and low cover

Collect complete co-located fractional-cover and purity truth at the intended footprint, including true zeros and low positives. Synthetic mixing weights are exact algebra, not a substitute for field fractions. Training has120 independent parents, not4,080 independent biological samples.

Cross sites and years with contemporary labels

Retain all valid repeats, acquisition/annotation dates and source identity. Record turnover and persistent identity linkage independently. Later annotations do not verify an earlier biological state. More regions and crossed years are needed to separate observation conditions from ecological change.

Keep scientific and computational uncertainties explicit

Identity and parent-bundle intervals condition on observed support. They are not guarantees over unseen regions or field cover. Model comparisons are exploratory outer evaluations; confidence sets and detection rules have costs. No current use is operationally validated.

Every primary result has a numerical source

Primary versioned packet

provenance/researcher-pft-information/2026-10-15/v1/ contains METHODS, RESULTS, LIMITATIONS, configuration/environment, the category-only label audit, editable report JSON/Markdown, figure manifests, numerical validation and canonical CSV/Parquet products.

Numerical products

researcher_pft_information_atlas.csv/.parquet:3,600 pair-coordinate rows. abundance_model_comparison.parquet: class/model/domain errors. tables/ contains every domain curve, class error, confusion matrix, synthetic truth/parent membership, prediction index, ambiguity analysis and paired bundle interval.

Historical evidence is preserved

Forty researcher-category permutation ensembles and prior researcher-only temporal class metrics are reused with explicit scope. Earlier dual-label reports/builders remain frozen for provenance and are no longer the primary biological interpretation. No source table was copied or historical output deleted.

Methods and validation

The report and publication figures rebuild from saved numerical products. Models, scalers, selection, parent roles, prediction math, simplex constraints and source hashes are checked separately. METHOD references include official scikit-learn grouped CV/PLS/boosting contracts and Duchi et al.2008, doi:10.1145/1390156.1390191 for simplex projection; inherited permutation/MESMA definitions remain linked in METHODS.

Researcher categories are the sole primary grouping. Statistical difference, recognition, known synthetic mixing weights and independent physical abundance remain distinct claims.